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openBURST: Real-time air surveillance simulation and analysis for active and passive sensors

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Summary

openBURST[1] is intended to provide the air surveillance sensor community with a framework for the development and testing of sensor coverage and real-time target detection analysis. openBURST focuses on the overall performance assessment of a sensor network. Besides static coverage diagram computations, openBURST facilitates real-time computation of sensor detections for simulated targets. This allows the statistical performance analysis of air picture generation in a given scenario. By providing a flexible and extendable framework to model and simulate active, passive, monostatic and multistatic sensors, openBURST provides the implementation of optimization and machine learning approaches for sensor portfolio performance optimization in air surveillance.

Statement of need

openBURST is a software project to design, deploy and test active/passive radar system networks for detecting simulated or recorded real world air traffic. It consists of decoupled software modules that can be replaced, extended or deployed independently for air surveillance sensor coverage and real-time detection computations. openBURST uses real-time communication between the distributed modules of the simulation framework, allowing for concurrent updates of target movements and sensor detections. Currently, openBURST supports coverage computation and real-time simulation of active radar and passive radar sensors (range-Doppler maps) using FM transmitters. The RF signal propagation, loss and terrain analysis tool Splat!, is extended for EM signal propagation computations with multi-core parallel processing and graphical user interfacing. openBURST uses openstreetmap data with openlayers for the interactive map. Terrain digital elevation data provided by GMTED10 is used for Line-of-Sight and propagation loss computations. By implementing a client-server architecture, openBURST enables browser based clients to remain data and implementation agnostic.

Sensor simulation for air surveillance is of considerable interest both in the civilian and military fields as it allows for performance assessments of arbitrary sensor networks in diverse scenarios of interest. Such a simulation tool should be seen as the combination of three subparts: air traffic simulation, sensor detection simulation and sensor data fusion.

In the opensource domain, several tools exist for the subpart of air traffic simulation. E.g. BlueSky (Hoekstra & Ellerbroek, 2016) presents a fully open-source and open-data approach to air traffic simulation, whereas the OpenSky network (Schäfer et al., 2014) is a non-profit community-based receiver network that collects air traffic surveillance data and makes it accessible to academic and institutional researchers. In the domain of air surveillance sensor

[1] BURST is an acronym for "Base Units Readiness Simulation Tool". Burst 1969 mASL is a mountain peak in the Emmentaler Alps of canton Berne, Switzerland. The opensourced version of the original BURST tool was named openBURST.

detection simulation, opensource tools for passive radar simulation such as PassiveRadarSim and for active radar simulation such as radarsimpy have been published. In the active radar domain, sophisticated opensource tools for very specific use cases such as for Synthetic Aperture Radar (SAR) RaySAR, or variants of generic wave propagation tools such as gprMAX also exist. Diverse data fusion, tracking and state estimation tools have also been opensourced such as Stonesoup (Thomas et al., 2017). However, currently no extendable real-time simulation suite for air traffic simulation and sensor detection for active/passive sensors that incorporates wave propagation modelling is available to the opensource community. openBURST is an attempt to fill this gap, proving the research and air surveillance simulation community with a general-purpose and modular tool, that provides a user-friendly browser based interface.

On contrary to a number of existing open source tools for detailed simulations of single sensors (either active or passive, monostatic or bistatic etc.), openBURST provides an extendable framework for diverse sensor and target simulations. By parallelizing computationally intensive steps and by providing a user-friendly browser based interface, openBURST considerably simplifies sensor network and target simulations. For example, openBURST can be used to dynamically change scenarios, sensor/transmitter attributes and targets for predicting passive/active radar performance and compare with real-world test measurements as needed by recent studies (Malanowski et al., 2022). openBURST provides a modular and extendable active/passive sensor network simulation framework, that can be used as the sensor model in diverse studies, such as sensor fusion performance (Fränken et al., 2022), sensor location optimization (Mathews et al., 2015; Yi et al., 2017), learning resource optimization (Mathews et al., 2022) or decision making (Mathews et al., 2008). In summary, openBURST facilitates new exciting scientific explorations of sensor network/fusion performance studies and sensor network performance benchmarking. Extensions with digital transmitters for passive radar and passive emitter tracking sensors are currently planned.

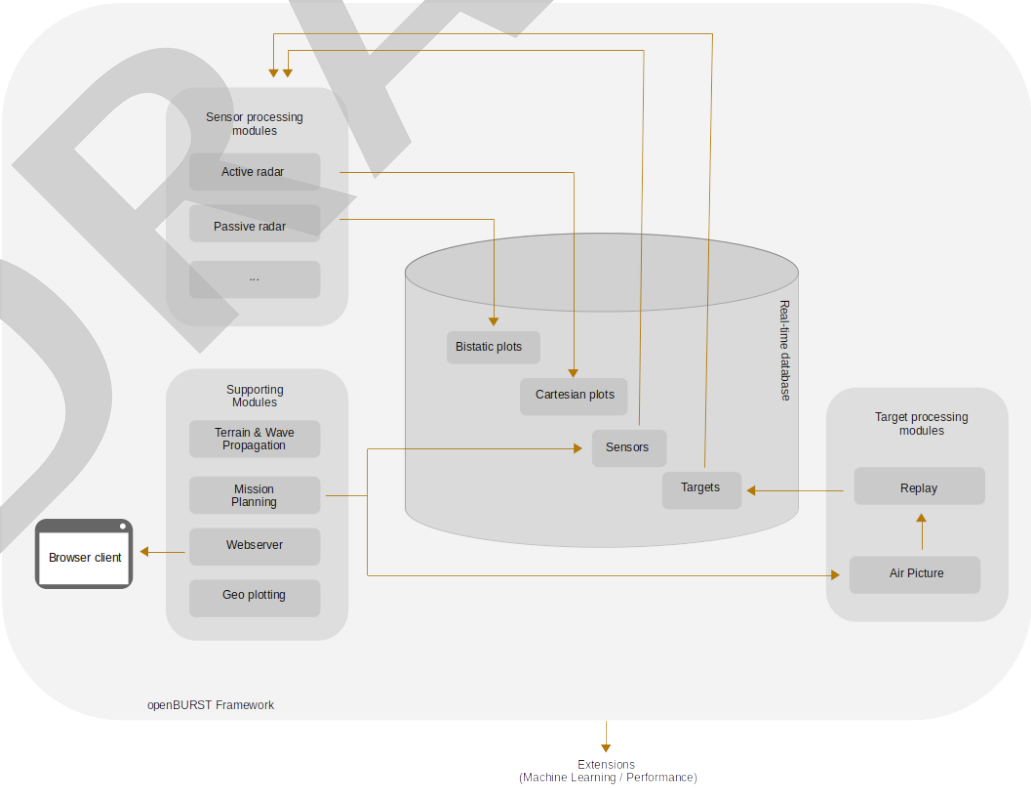

Figure 1: openBURST framework with its component modules and real-time database.

Results Examples

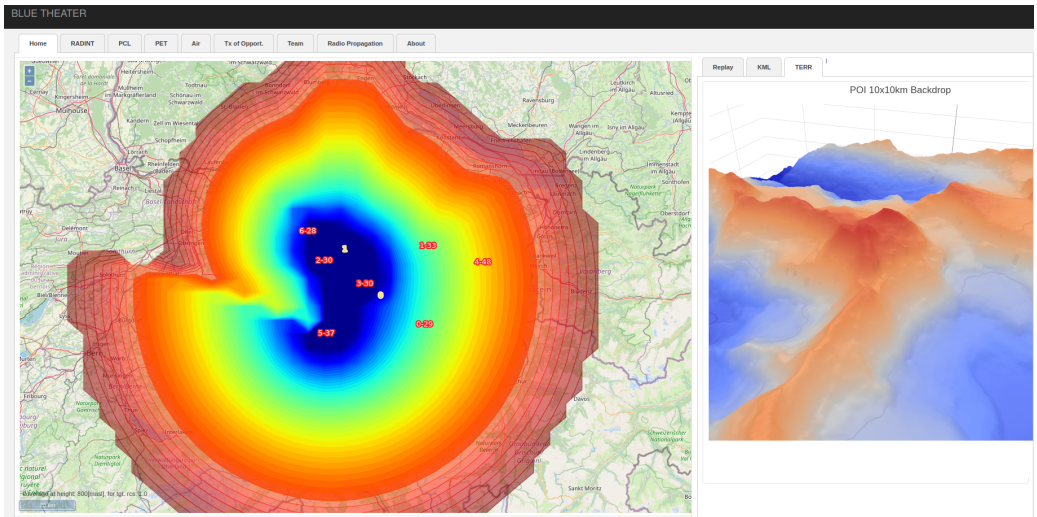

Figure 2: Example of passive radar coverage computation. Left: with numbered receivers and transmitters of opportunity, right: terrain backdrop of the sensor position.

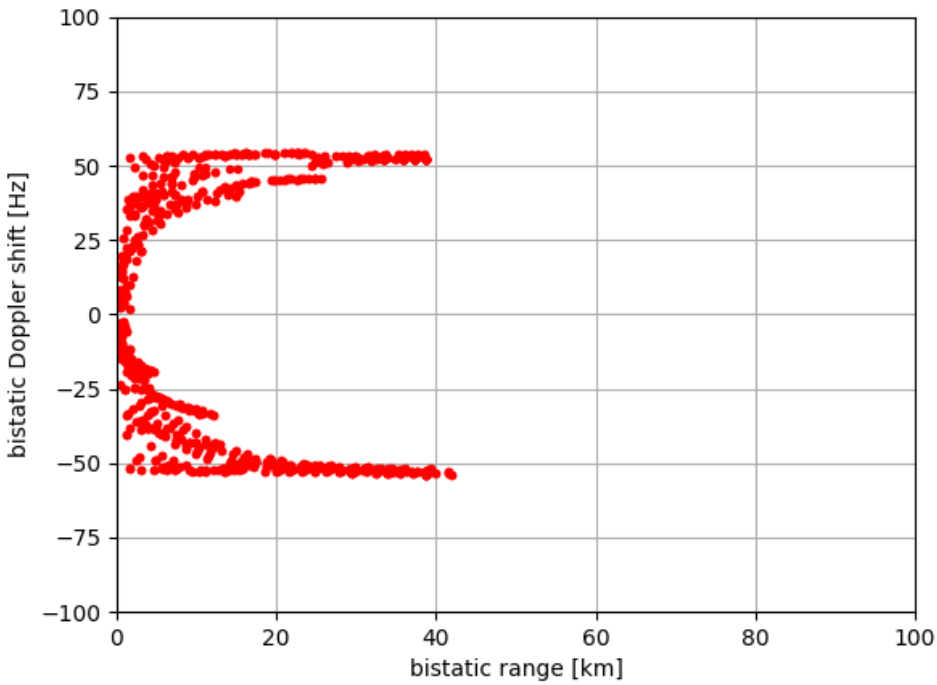

Figure 3: Example of passive radar real-time range-Doppler detections for a given target scenario of a closely flying formation of aerial vehicles.

Publications and Projects

Versions of openBURST have been used in a number of studies for passive radar location optimization, sensor fusion, learning resource allocation in active and passive sensor networks (Mathews et al., 2015, 2022; Mousel, 2017; Studer et al., 2022).

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